

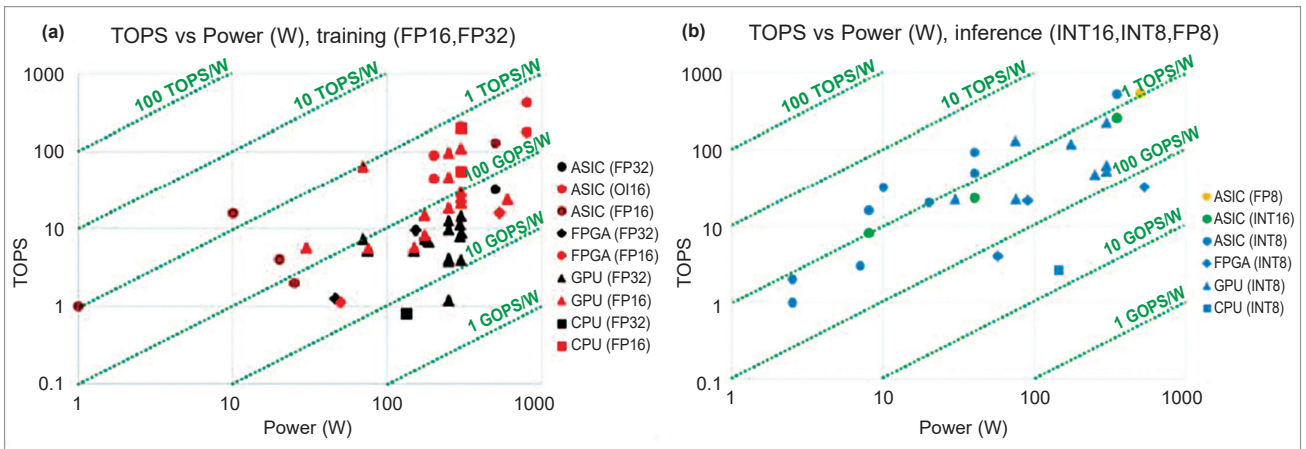


Fig. 3: TOPS (data points) and the resulting TOPS/W (assessed against the green dotted lines) as a function of reported system power in (a) the training and (b) the inference

operations to ensure our hardware effectively handles AI applications.

Initially, the prevailing wisdom among system designers was that expanding compute power primarily involved ramping up the number of CPUs and cores to achieve maximum performance. However, contemporary benchmarking efforts have revealed that this approach does not yield a direct and flawless linear scaling. Increasing the number of cores, while beneficial to a certain extent, encounters limitations related to power consumption and other systemic architectural constraints.

To meet the demands of AI applications, it is imperative that we delve into other facets of system design. This shift in perspective is crucial to facilitate the seamless transition of our hardware to accommodate diverse use cases. For many of us in this field, this shift in thinking is of personal interest. It brings into focus the critical importance of application benchmarking in assessing the overall system design.

The quest for optimal solutions extends beyond mere processing power and touches upon the intricacies of power efficiency, architectural design, and the diverse needs of AI applications. In this journey, time to market remains a significant consideration.

Complex application benchmarking has become an integral part of our strategy as we strive to bridge the gap between hardware capabilities and the evolving demands of AI applications. Through these efforts, we aim to ensure that our hardware is not just keeping pace but leading the way in the dynamic realm of AI technology.

Customisation and domain-specific chipsets

Traditionally, semiconductor designers aimed to increase compute power by adding more general-purpose CPUs and cores. However, as benchmarking has shown, there are limits to this approach, constrained by factors like power consumption and system architectural limitations. This realisation has led to a shift towards domain-specific chipsets.

Today, chipsets can be designed to serve specific domains or applications. Rather than a one-size-fits-all approach, these domain-specific chipsets can be tailored to the unique requirements of different applications, such as web browsing, minimal applications, multimedia, image recognition, and more. This customisation allows chipsets to be more efficient and cost-effective, meeting the specific needs of each application.

One of the most exciting develop-

ments in semiconductor design is the emergence of software-defined hardware. This paradigm shift recognises that while general-purpose chipsets have their place, the future lies in custom, adaptable chips designed for specific tasks.

Software-defined hardware allows chip designers to create modules that can be activated or deactivated depending on the application's requirements. This flexibility ensures that the chip remains power-efficient and capable of delivering high performance for the intended task. It is a revolutionary approach that addresses the dynamic needs of the modern AI landscape.

Beyond chip design, AI is making significant inroads into the entire semiconductor lifecycle. McKinsey's research suggests that AI can play a crucial role in various stages, from capital allocation and product portfolio definition to chip design and supply chain management.

AI-driven automation promises to streamline processes, improve operational efficiencies, and reduce costs. For example, Google's open chip architecture leverages AI to design chipsets, demonstrating the potential of AI in semiconductor design. Additionally, AI can optimise supply chain operations, mitigating issues like the